American Computer Science League

2021-2022 ● Contest 1: Short Problem Solutions ● Elementary Division

1. Computer Number Systems

To convert 147_8 to base 10, take powers of 8 times each digit in the number:

$7 + 4 * 8 + 1 * 64 = 7 + 32 + 64 = 103_{10}$

D. 103

2. Computer Number Systems

Converting each to binary yields the following so $B14_{16}$ has the fewest 1s:

$9A7_{16} = 100110100111 = 7$ digits that are 1
$B14_{16} = 101100010100 = 5$ digits that are 1
$7431_8 = 111100011001 = 7$ digits that are 1
$4265_8 = 100010110101 = 6$ digits that are 1

B. B14

3. Computer Number Systems

To find the year $3F7_{16}$ years from now which is $84A_{16}$, add them together;

A + 7 = 10 + 7 = 17 so keep the 1 and carry 16.
4 + F + 1 (carried) = 4 + 15 + 1 = 20 so keep the 4 and carry 16.
8 + 3 + 1 (carried) = 12 = C_{16} so it's $C41_{16}$.

E. C41

4. Computer Number Systems

To find the largest, you must convert all of them to either base 8 or base 16:

$111001101010_2 = E6A_{16}$
$E60_{16}$ and $E69_{16}$ are less than that.
7147_8 is less than 7150_8 which is 111 001 101 000 = 1110 0110 1000 which is $E68_{16}$. It is also less than $E6A_{16}$.

A. 111001101010

5. Computer Number Systems

The most dominant component in the hexadecimal color "#9FA0B1" is B1 which is the blue component.
In base 10, B1 = 1 + B(11) * 16 = 1 + 176 = 177_{10}
The RGB color using base 10 numbers (159,160,177) is Santa Grey.

B. 177